EE2004: Digital Signal Processing

Tutorial 3

(February 12, 2026)

1. Find the z-transform of the following sequences and state clearly the RoC in each case: (a) $(0.8)^n\,u[n] + 2^n\,u[-n-1]$, (b) $2^n\,u[n] - 3^n\,u[-n-1]$, (c) $(0.8)^n\,u[n] + (0.9)^n\,u[-n-1]$, (d) $\left[(0.8)^n + 3\,(0.4)^n\right]u[-n-1]$, (e) $\left[(0.8)^n + 3\,(0.4)^n\right]u[n]$, and (f) $(0.5)^{|n|}$.

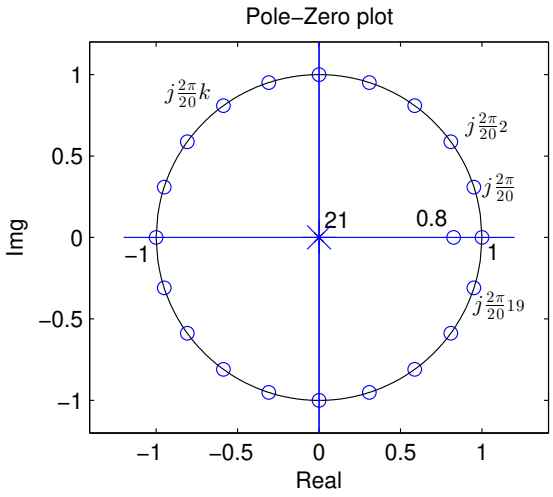
2. Let $x[n] = e^{j\omega_0 n}\,u[n]$. Find $X(z)$ and its corresponding RoC.

3. Let $x[n] = \cos(\omega_0 n)\,u[n]$. Find $X(z)$ and its corresponding RoC. Repeat for $x[n] = \sin(\omega_0 n)\,u[n]$. In each case, plot the pole-zero plot of $X(z)$.

4. Let $H(z) = \dfrac{(e^{-2}-2)z}{(z-e^{-2})(z-2)}$. List all possible RoCs corresponding to the given $H(z)$. Also state the nature of the corresponding time-domain sequence for each case, i.e., left-sided, right-sided, or two-sided.

5. Find $X(z)$ when $x[n]$ equals (a) $\delta[n]$, (b) $\delta[n-n_0]$ $(n_0 > 0)$, and (c) $\delta[n+n_0]$ $(n_0 > 0)$. What is the RoC in each case?

6. Find $X(z)$ when $x[n] = u[n]$ (the unit step sequence). What is its RoC?

7. 

   Consider the pole-zero plot shown on the left. Find the corresponding $H(z)$ whose pole-zero plot is consistent with the given information.

8. Let $x[n] = r^n\,\dfrac{\sin[(n+1)\theta]}{\sin\theta}\,u[n]$. Find $X(z)$ and indicate the RoC. You should not assume the z-transform of either $\cos(\omega_0 n)\,u[n]$ or $\sin(\omega_0 n)\,u[n]$ while deriving the final expression.

9. **Computer assignment** Create the signal $x[n] = \cos\left(\dfrac{n\pi}{5}\right)$ for $0 \le n \le 19$. Delay this signal by 2.5 samples as follows: (i) multiply the signal's DTFT (implemented using Matlab's `fft` command) by $\exp(-j\,2.5\omega)$ (after discretizing the frequency variable) over the the interval $[0, \pi]$; (ii) for the samples in the range $(\pi, 2\pi)$, fill them using the time-reversed, complex-conjugated samples taken from the interval $[0, \pi]$; (iii) find

the inverse transform to obtain a sequence that is 20 samples in length (it must be real-valued, and any imaginary part present must be very small and should be zeroed out). Plot this alongside of $y[n] = \cos\left(\frac{n\pi}{5} - \frac{\pi}{2}\right)$. Also compare this with `delayseq(x, 2.5)`.

10. **Computer assignment** Load the speech waveform contained in the file `f0125s.wav` (look up Matlab's `audioread` command). Add a delayed version of the speech to itself. Consider delays in the range 50–100 ms. Note the difference in sound quality. Increase the delay and observe what happens for values much greater than 100 ms.

11. **Computer assignment** Load the speech file `f0103s.wav`. Pass it through a block that introduces a *time-varying* delay. That is, this block produces the output is $x[n - d_n]$. Choose `d = 20 + 20*sin(2*pi*5*n/Fs)`. Implement this block using Matlab's `delayseq` command, which has the capability to introduce fractional delays. A *vibrato* effect is imparted to the output. Experiment with variable delay parameters to see how they influence the vibrato effect.

12. **Computer assignment** Load the speech file `f0126s.wav`. Consider the following two different variable delay values, i.e., `d1 = 200 + 32*sin(2*pi*2*n/Fs)` and `d2 = 200 + 50*sin(2*pi*3*n/Fs)`. Pass the original signal through each of these two variable delay blocks. Listen to each output and compare it with the original. In particular, are you able to note the changes in the words "along" and "one mile lane"?; are there any other noticeable differences? Next, add the two outputs to the original and listen to combined signal. This produces a *chorus* effect. Vary the parameters of the delay blocks and observe how they change the sound quality.